

Introduction to Machine Learning

What is Machine Learning?

Machine Learning (ML) is a subset of Artificial Intelligence that enables systems to learn from data and improve performance without being explicitly programmed.

Types of Machine Learning

1. Supervised Learning: Uses labeled data. 2. Unsupervised Learning: Finds patterns in unlabeled data. 3. Reinforcement Learning: Learns through rewards and penalties.

Common Algorithms

Algorithm	Use Case
Linear Regression	Predict continuous values
Decision Trees	Classification tasks
K-Means	Clustering
Neural Networks	Deep learning applications

Applications of ML

Machine Learning is widely used in recommendation systems, fraud detection, image recognition, natural language processing, and autonomous vehicles.